

Stability of Weakly Unique Hahn–Banach Extensions under Bochner L^p and $C(K)$ Constructions

Candidate full solution – run `fa_banach_001`, lane 14

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Candidate full result; likely valid. Under the standing Radon–Nikodym hypothesis in Daptari–Paul, both converse questions after Theorems 2.2.2 and 2.2.3 have affirmative answers. Thus property-(wU) passes both to Bochner L^p spaces, $1 < p < \infty$, and to $C(K)$ spaces.

Source question

Daptari and Paul [1] work over real Banach spaces. In the paragraph containing Theorems 2.2.2 and 2.2.3 they fix a probability space (Ω, Σ, μ) , a closed subspace $Y \subset X$, and assume that X^* has the Radon–Nikodym property (RNP). They prove the implications from the function-space property back to $Y \subset X$, then ask whether their converses hold. The exact source passage is on source PDF page 7 (source lines 278–293).

We now discuss the cases of vector-valued functions. A contra-positive argument leads to the following hypothesis. Let (Ω, Σ, μ) be a probability space and Y be a subspace of X such that X^* has the Radon–Nikodým property (see [DU] pg. 61] for the definition). We denote this property by RNP in the remainder of this paper.

Theorem 2.2.2. *If $L_p(\mu, Y)$ has property-(wU) in $L_p(\mu, X)$, then Y has property-(wU) in X , where $1 < p < \infty$.*

Furthermore, we have the following.

Theorem 2.2.3. *Let Y be a subspace of X , and let X^* has the RNP. If $C(K, Y)$ has property-(wU) in $C(K, X)$, then Y has property-(wU) in X .*

We do not know the validity of the converse of Theorem [2.2.2, 2.2.3]

Figure 1: The standing RNP hypothesis, Theorems 2.2.2–2.2.3, and the open converse question in [1].

Transcription. “We do not know the validity of the converse of Theorem 2.2.2, 2.2.3.”

Definitions

A closed subspace Y has *property-(wU)* in X if every norm-attaining $y^* \in Y^*$ has a unique norm-preserving Hahn–Banach extension to X . Here $L^p(\mu; X)$ is the Bochner space and $C(K; X)$ is the space of continuous X -valued functions on a compact Hausdorff space K .

Proof intuition

The mechanism is pointwise. A norm-attaining functional on a Bochner space has a dual density $g(\omega)$. Equality in the two inequalities used to estimate its value forces $g(\omega)$ itself to be norm-attaining almost everywhere. If $h(\omega)$ represents a norm-preserving extension, restriction gives $\|h(\omega)\| \geq \|g(\omega)\|$, while equality of the global norms forces pointwise equality. Property-(wU) then identifies all extension densities.

For $C(K; X)$, the same argument is performed with representing vector measures. Restriction decreases total variation. Since a norm-preserving extension has the same total mass, its variation measure must equal that of the original functional. RNP densities then reduce uniqueness to property-(wU) pointwise.

Result

Theorem 1 (Both converse questions). *Let X be a real Banach space, let $Y \subset X$ be a closed subspace, and assume that X^* has the RNP.*

(i) *If (Ω, Σ, μ) is a probability space and $1 < p < \infty$, then*

$$Y \text{ has property-(wU) in } X \iff L^p(\mu; Y) \text{ has property-(wU) in } L^p(\mu; X).$$

(ii) *If K is a nonempty compact Hausdorff space, then*

$$Y \text{ has property-(wU) in } X \iff C(K; Y) \text{ has property-(wU) in } C(K; X).$$

Proof. We first record the easy implications from right to left, which are the directions proved in [1, Theorems 2.2.2–2.2.3]. If property-(wU) fails for $Y \subset X$, choose a norm-one $y^* \in Y^*$, a point $y_0 \in S_Y$ with $y^*(y_0) = 1$, and two distinct norm-one extensions $x_1^*, x_2^* \in X^*$.

For the L^p assertion, the functionals

$$F_j(f) = \int_{\Omega} x_j^*(f(\omega)) d\mu(\omega), \quad j = 1, 2,$$

are distinct norm-one extensions of the same functional on $L^p(\mu; Y)$ and attain their norm at the constant function y_0 . For $C(K)$, fix $t_0 \in K$ and use $F_j(f) = x_j^*(f(t_0))$. Thus the function-space property implies the base-space property.

It remains to prove the two converse implications.

Bochner L^p spaces. Let $q = p/(p-1)$. The RNP passes to the quotient Y^* of X^* , so the standard Bochner duality theorem [2] gives the isometric representations

$$(L^p(\mu; X))^* = L^q(\mu; X^*), \quad (L^p(\mu; Y))^* = L^q(\mu; Y^*).$$

Assume that Y has property-(wU) in X . Let a nonzero functional on $L^p(\mu; Y)$ be represented by $g \in L^q(\mu; Y^*)$ and attain its norm at $f \in S_{L^p(\mu; Y)}$. Replacing f by $-f$ if needed, we have

$$\begin{aligned}\|g\|_q &= \int_{\Omega} g(\omega)(f(\omega)) d\mu(\omega) \\ &\leq \int_{\Omega} \|g(\omega)\| \|f(\omega)\| d\mu(\omega) \leq \|g\|_q \|f\|_p = \|g\|_q.\end{aligned}$$

Both inequalities are equalities. The first gives

$$g(\omega)(f(\omega)) = \|g(\omega)\| \|f(\omega)\| \quad \text{for a.e. } \omega.$$

Equality in scalar Holder, with $1 < p, q < \infty$, shows that $f(\omega) \neq 0$ wherever $g(\omega) \neq 0$, up to a null set. Consequently $g(\omega)$ is norm-attaining for almost every ω on which it is nonzero.

Now let two norm-preserving extensions be represented by $h_1, h_2 \in L^q(\mu; X^*)$. If $r : X^* \rightarrow Y^*$ is restriction, uniqueness of the Bochner representation gives

$$r(h_j(\omega)) = g(\omega) \quad \text{a.e.,} \quad j = 1, 2.$$

Hence $\|h_j(\omega)\| \geq \|g(\omega)\|$ almost everywhere. But

$$\int \|h_j\|^q d\mu = \|h_j\|_q^q = \|g\|_q^q = \int \|g\|^q d\mu,$$

so $\|h_j(\omega)\| = \|g(\omega)\|$ almost everywhere. On the nonzero set, $h_j(\omega)$ is therefore a norm-preserving extension of the norm-attaining functional $g(\omega)$; property-(wU) yields $h_1(\omega) = h_2(\omega)$. On the zero set, norm equality forces both densities to vanish. Thus the two global extensions coincide. The zero functional has only the zero norm-preserving extension, completing (i).

C(K) spaces. Again assume that Y has property-(wU) in X . By the vector-measure representation theorem [2], let a nonzero norm-attaining functional on $C(K; Y)$ correspond to a regular Y^* -valued measure m of bounded variation. Since Y^* has the RNP, write

$$m(A) = \int_A g(t) d|m|(t), \quad \|g(t)\| = 1 \quad |m|\text{-a.e.}$$

Let $f \in S_{C(K; Y)}$ be chosen with the functional taking the value $\|m\| = |m|(K)$ at f . Then

$$|m|(K) = \int_K g(t)(f(t)) d|m|(t) \leq \int_K \|f(t)\| d|m|(t) \leq |m|(K).$$

Thus $g(t)(f(t)) = 1$ for $|m|$ -almost every t . In particular, $g(t)$ is a norm-attaining member of Y^* almost everywhere.

Let n be the regular X^* -valued measure representing any norm-preserving extension. Equality on $C(K; Y)$ gives $r \circ n = m$. Therefore, for every Borel set A ,

$$|m|(A) \leq |n|(A).$$

The total masses are the functional norms, so $|n|(K) = |m|(K)$. The positive measure $|n| - |m|$ has total mass zero, whence

$$|n| = |m|.$$

The RNP of X^* now gives a Bochner density

$$n(A) = \int_A h(t) d|m|(t), \quad \|h(t)\| = 1 \quad |m|\text{-a.e.}$$

The identity $r \circ n = m$ implies $r(h(t)) = g(t)$ almost everywhere. Thus $h(t)$ is a norm-preserving extension of the norm-attaining $g(t)$, and property-(wU) determines it uniquely almost everywhere. Any two representing extension measures therefore coincide. This proves (ii). $\square$

Verification and scope

The proof retains every standing hypothesis from the source. It uses no measurable selection: the densities of the extensions are already supplied by the dual and vector-measure representation theorems. The zero functional, null sets, and both directions of each equivalence are handled explicitly.

The result is not asserted at $p = 1$ or $p = \infty$, without the RNP hypothesis, or over complex scalars. The last case should admit a phase-normalized variant, but it is outside the source’s stated real setting.

Novelty check

Bounded searches on 9 August 2026 used the exact title and arXiv id, the exact open-question sentence, ‘property-(wU)’ together with $L^p(\mu; Y)$ and $C(K; Y)$, and ‘weak Hahn–Banach extension property’ with vector-valued function spaces. They found the source and mirrors, but no later paper claiming either converse. This is a bounded search, not a definitive priority claim.

Human-review recommendation

Review the two standard representation inputs and, in the $C(K)$ proof, the step $|m| \leq |n|$ plus equality of total masses. Once those are accepted, the remaining argument is pointwise and direct.

References

- [1] S. Daptari and T. Paul, *Uniqueness of Hahn–Banach extensions and some of its variants*, arXiv:2202.00947, 2022.
- [2] J. Diestel and J. J. Uhl, Jr., *Vector Measures*, Mathematical Surveys 15, American Mathematical Society, Providence, 1977.